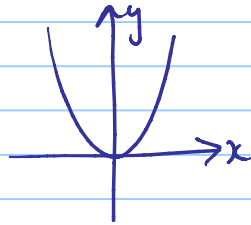


A function is a one-to-one function, if it never takes on the same value twice.

If it is one-to-one and $f(x_1) = f(x_2)$, then $x_1 = x_2$.

Examples:



$y = x^2$ is not one-to-one.



$y = e^x$ is a one-to-one function.

ONE-TO-ONE FUNCTIONS HAVE INVERSES.

Definition: p. 55

If f is one-to-one with domain A and range B , then its inverse function f^{-1} has domain B and range A and

$$f^{-1}(y) = x \iff y = f(x).$$

$$D_{f^{-1}} = R_f \text{ and } R_{f^{-1}} = D_f.$$

Cancellation equations:

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x \quad \forall x \in A$$

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = x \quad \forall x \in B.$$

E.g. if $f(x) = x^3$, then $f^{-1}(x) = x^{1/3}$ and

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = f(x^{1/3}) = (x^{1/3})^3 = x$$

$$\text{and } (f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

NB 3 steps to find an inverse of function $f(x)$.

① Write $y = f(x)$

② Interchange x and y

③ Solve this equation to find y in terms of x , then $y = f^{-1}(x)$.

Examples:

1) Do example 4 on p. 57.

② Find the inverse of $f(x) = \sqrt{-1-x}$. Sketch f and f^{-1} .

$$\begin{aligned}y &= \sqrt{-1-x} \\ \text{Let } x &= \sqrt{1-y} \\ \Rightarrow x^2 &= 1-y \\ \Rightarrow -y &= x^2 + 1 \\ \text{and } y &= -x^2 - 1. \\ \Rightarrow f^{-1}(x) &= -x^2 - 1.\end{aligned}$$

$$\begin{aligned}f(x) &= \sqrt{-1-x} \\ -1-x &\geq 0 \\ \Rightarrow -1 &\geq x \Rightarrow x \leq -1 \\ D_f &= (-\infty, -1] = R_{f^{-1}} \\ R_f &= [0, \infty) = D_{f^{-1}}\end{aligned}$$

